

FedMut: Federated Prediction of Combinatorial Mutation Effects Using Pre-trained Protein Language Models

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Abstract

Predicting the functional effects of combinatorial mutations is a critical challenge in protein engineering and evolutionary biology. While Deep Mutational Scanning (DMS) provides ground-truth fitness landscapes, the data is often siloed across different institutions and sparse due to the vast combinatorial space of mutations. We present FedMut, a federated learning framework that enables collaborative training of DMS score prediction models across distributed datasets without sharing raw sequence data. Our approach leverages pre-trained protein language models from NVIDIA BioNeMo as feature extractors, combined with a trainable regression head that is optimized using Federated Averaging (FedAvg). We evaluate our method on the ProteinGym Substitution Benchmark, comprising approximately 2.4 million missense variants across 217 DMS assays, partitioned into four federated clients based on biological domain (Human, Virus, Prokaryote, Eukaryote). By freezing the protein language model backbone and only training the prediction head in a federated manner, we hope to reduce communication overhead while maintaining competitive predictive performance. Our framework addresses key challenges in cross-institutional collaboration for protein fitness prediction, enabling privacy-preserving model training that benefits from diverse biological datasets.

1 Introduction

Protein engineering and evolutionary biology rely on understanding how mutations affect protein function and stability. Deep Mutational Scanning (DMS) experiments provide comprehensive fitness landscapes by systematically quantifying the effects of mutations on protein function (Fowler and Fields, 2014). However, the combinatorial space of mutations is vast, rendering exhaustive experimental characterization infeasible. Consequently, computational methods that accurately predict DMS scores from protein sequences are essential for guiding protein design and elucidating evolutionary constraints.

Recent advances in protein language models, such as ESM-2 (Lin et al., 2023) and models available through NVIDIA BioNeMo (?), have demonstrated remarkable capabilities in learning meaningful representations of protein sequences from large-scale unlabeled data. These pre-trained models can be fine-tuned or used as feature extractors for downstream tasks including fitness prediction (Meier et al., 2021, Notin et al., 2023).

Despite these advances, a significant challenge remains: DMS data is often distributed across different institutions—hospitals, academic laboratories, and industry—each with their own proprietary or sensitive datasets. Traditional centralized training approaches require sharing raw sequence data, which raises privacy concerns and may be infeasible due to regulatory constraints or competitive interests. Federated learning (FL) offers a promising solution by enabling collaborative model training without data sharing (McMahan et al., 2017).

In this work, we introduce FedMut, a federated learning framework for predicting DMS scores that addresses both the sparsity and siloed nature of mutation effect data. By leveraging pre-trained protein language models as frozen feature extractors, we minimize computational and communication costs while enabling privacy-preserving collaborative training across multiple institutions. We demonstrate our approach on the ProteinGym Substitution Benchmark, simulating a realistic cross-institutional collaboration scenario with four distinct client nodes representing different biological domains.

2 Methods

2.1 Dataset

We utilize the ProteinGym Substitution Benchmark (Notin et al., 2023), a standardized collection of approximately 2.4 million missense variants across 217 DMS assays. This benchmark provides consistent input formats and evaluation protocols, making it ideal for federated learning simulations. Each dataset in our simulation corresponds to a single DMS assay and adheres to the following schema:

- **mutant**: A colon-separated string describing amino acid changes relative to the reference sequence (e.g., **A1P:D2N** indicates Alanine at position 1 $\rightarrow$ Proline, and Aspartic Acid at position 2 $\rightarrow$ Asparagine).
- **mutated_sequence**: The full, explicit amino acid sequence of the variant protein, serving as the primary input for the protein language model feature extractor.
- **DMS_score**: The experimental ground-truth value, where higher scores indicate higher fitness or functional retention of the mutated protein. This serves as the regression target.
- **DMS_score_bin**: A binarized classification label based on assay-specific fitness cutoffs (1 = fit/pathogenic; 0 = not fit/benign).

In addition to sequence data, we leverage ProteinGym metadata including UniProt IDs, taxonomic classification (Taxon), and Multiple Sequence Alignment (MSA) depth to partition and balance data across federated clients.

2.2 Federated Client Simulation

To simulate a realistic cross-institutional collaboration, we partition the ProteinGym Substitution Benchmark into four distinct client nodes based on biological domain (Taxon). We aggregate all available assays corresponding to a specific taxon into a single client node, ensuring that each client possesses a comprehensive and heterogeneous local dataset rather than a single representative protein. The client assignments are as follows:

2.3 Model Architecture

We employ a transfer learning approach with a frozen backbone and a trainable regression head. This design choice minimizes communication overhead and computational requirements on edge clients while leveraging powerful pre-trained representations.

Table 1: Federated client nodes and their simulation scenarios

Node	Client Type	Simulation Scenario
Client 1	Human	Clinical Hospital / Oncology
Client 2	Virus	Virology Lab / Pandemic Prep
Client 3	Prokaryote	Antibiotic Resistance Lab
Client 4	Eukaryote	Academic Bio-Foundry

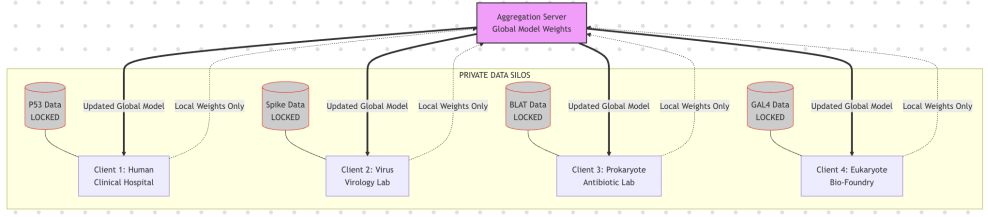


Figure 1: Distribution of DMS assays across federated client nodes based on biological domain.

2.3.1 Frozen Backbone (BioNeMo)

We use a pre-trained Protein Language Model (e.g., ESM-2 or MegaMolBART via NVIDIA BioNeMo) as the encoder. The model takes as input the amino acid sequence of the mutant protein and outputs per-residue or whole-sequence embeddings. All weights in this encoder are frozen during federated training to:

1. Reduce communication overhead (only prediction head weights are transmitted)
2. Minimize computational requirements on edge clients
3. Leverage representations learned from massive protein databases

2.3.2 Trainable Prediction Head

The prediction head consists of two components:

- **Pooling Layer:** Aggregates the sequence embedding into a fixed-size vector using mean pooling or the <CLS> token representation.
- **Regression MLP:** A multi-layer perceptron stacked on top of the pooled embeddings. The architecture details (number of layers, hidden dimensions, activation functions) are to be determined during hyperparameter optimization.

The output is a single scalar value representing the predicted DMS score.

2.4 Federated Learning Protocol

We employ the Federated Averaging (FedAvg) algorithm (McMahan et al., 2017) for collaborative training. The training process proceeds as follows:

1. **Initialization:** The central server initializes the weights of the prediction head and distributes them to all four clients. The BioNeMo backbone is pre-loaded on all clients.

DMS_index	DMS_id	DMS_filename	UniProt_ID	taxon	source_organism	target_seq
DMS_sub_0	A0A140D2T1_ZIKV_Sourisseau_2019	A0A140D2T1_ZIKV_Sourisseau_2019.csv	A0A140D2T1_ZIKV	Virus	Zika virus (ZIKV)	MKNPKKKSGGFRIVK
DMS_sub_1	A0A192B1T2_9HIV1_Haddox_2018	A0A192B1T2_9HIV1_Haddox_2018.csv	A0A192B1T2_9HIV1	Virus	Human immunodeficiency virus 1	MRVKGIQMNSQHLLF
DMS_sub_2	A0A19GEU1_NEIME_Kennouche_2019	A0A19GEU1_NEIME_Kennouche_2019.csv	A0A19GEU1_NEIME	Prokaryote	Neisseria meningitidis	FTLIELMVAINGILAAI
DMS_sub_3	A0A247D711_LISMN_Stadelmann_2021	A0A247D711_LISMN_Stadelmann_2021.csv	A0A247D711_LISMN	Prokaryote	Listeria monocytogenes	MININDUREIKNDYIT
DMS_sub_4	A0A2Z5U3Z0_9INFA_Doud_2016	A0A2Z5U3Z0_9INFA_Doud_2016.csv	A0A2Z5U3Z0_9INFA	Virus	Influenza A virus (A/WSN/1933(H1N1))	MKAKLLVLLYAFVATD
DMS_sub_5	A0A2Z5U3Z0_9INFA_Wu_2014	A0A2Z5U3Z0_9INFA_Wu_2014.csv	A0A2Z5U3Z0_9INFA	Virus	Influenza A virus (A/WSN/1933(H1N1))	MKAKLLVLLYAFVATD
DMS_sub_6	A4_HUMAN_Seuma_2022	A4_HUMAN_Seuma_2022.csv	A4_HUMAN	Human	Homo sapiens	MLPGLALLLLAAWTAI
DMS_sub_7	A4D664_9INFA_Soh_2019	A4D664_9INFA_Soh_2019.csv	A4D664_9INFA	Virus	Influenza A virus (A/green-winged teal/Ohio)	MERIKELRDLMSQSR
DMS_sub_8	A4GRB6_PSEAI_Chen_2020	A4GRB6_PSEAI_Chen_2020.csv	A4GRB6_PSEAI	Prokaryote	Pseudomonas aeruginosa	MFKLLSKLLVYLTAIA
DMS_sub_9	AACC1_PSEAI_Dandage_2018	AACC1_PSEAI_Dandage_2018.csv	AACC1_PSEAI	Prokaryote	Pseudomonas aeruginosa	MLRSSNDVDTQQGSR
DMS_sub_10	ACE2_HUMAN_Chan_2020	ACE2_HUMAN_Chan_2020.csv	ACE2_HUMAN	Human	Homo sapiens	MSSSSWLLLSLWAVTI

Figure 2: Overview of the ProteinGym dataset structure and data schema.

2. **Local Training:** Each client trains the prediction head on their local data for a specified number of epochs. The loss function is Mean Squared Error (MSE) between predicted and actual DMS scores, optimized using AdamW.
3. **Aggregation:** Clients send only the updated weights of the prediction head back to the server. The server performs weighted averaging of these weights to create a new global model, where weights are proportional to the number of training samples per client.
4. **Distribution:** The updated global model is sent back to clients for the next round of federated training.

This process repeats for a predetermined number of communication rounds.

3 Results

3.1 Evaluation Metrics

We evaluate model performance using Spearman’s rank correlation coefficient between predicted and ground-truth DMS scores, which is robust to non-linear relationships and commonly used in protein fitness prediction benchmarks (Notin et al., 2023). Additional metrics including Pearson correlation, mean squared error, and mean absolute error will be reported.

3.2 Baseline Comparisons

We compare our federated learning approach (FedMut) against several baselines:

- **Local Training Only:** Each client trains independently on their local data without federated aggregation.
- **Centralized Training:** A baseline where all data is pooled and trained centrally (simulating the ideal but often infeasible scenario).
- **Pre-trained Model Only:** Using the frozen backbone without fine-tuning the prediction head.

3.3 Performance Results

After [X] rounds of federated training, the global model performance will be evaluated on held-out test sets for each client. The results will be reported in the following format:

Table 2: Model performance comparison across federated clients

Model	Client 1 (Human)	Client 2 (Virus)	Client 3 (Prokaryote)	Client 4 (Eukaryote)	Average
Local Training Only	TBD	TBD	TBD	TBD	TBD
Federated (FedMut)	TBD	TBD	TBD	TBD	TBD

3.4 Privacy and Communication Efficiency

We will analyze the communication overhead of our federated approach compared to centralized training, demonstrating the efficiency gains from freezing the backbone. Additionally, we will discuss the privacy guarantees provided by the federated learning framework.

4 Discussion

Our FedMut framework addresses critical challenges in collaborative protein fitness prediction by enabling privacy-preserving model training across distributed datasets. The use of frozen pre-trained protein language models as feature extractors provides a practical solution that balances performance, computational efficiency, and communication overhead.

The federated learning approach is particularly well-suited for the protein engineering domain, where data is often siloed across institutions due to proprietary interests, regulatory constraints, or competitive advantages. By allowing multiple institutions to collaboratively improve a shared model without sharing raw sequence data, FedMut enables the scientific community to leverage diverse biological datasets while respecting data privacy and ownership.

The partitioning of data by biological domain (Human, Virus, Prokaryote, Eukaryote) creates a realistic non-IID (non-independent and identically distributed) scenario that is common in federated learning applications. This heterogeneity tests the robustness of the FedAvg algorithm and may reveal opportunities for domain-specific adaptations or personalized federated learning approaches.

4.1 Future Directions

Several directions for future work are promising:

1. **Architecture Optimization:** Systematic hyperparameter search for the regression MLP architecture, including layer depth, hidden dimensions, dropout rates, and activation functions.
2. **Advanced Federated Algorithms:** Exploration of federated learning algorithms beyond FedAvg, such as FedProx (Li et al., 2020) or SCAFFOLD (Karimireddy et al., 2020), which may better handle the non-IID data distribution across biological domains.
3. **Multi-task Learning:** Extending the framework to simultaneously predict both continuous DMS scores and binary fitness classifications, potentially improving generalization.
4. **Real-world Deployment:** Validation of the framework with actual distributed institutions, addressing practical challenges such as network latency, client availability, and data quality heterogeneity.
5. **Interpretability:** Incorporating attention visualization or gradient-based methods to understand which sequence regions contribute most to fitness predictions, providing biological insights alongside predictions.

6. **Combinatorial Mutation Modeling:** Extending beyond single and double mutations to more complex combinatorial mutation patterns, potentially using graph neural networks or attention mechanisms to model interactions between mutation sites.

5 Conclusion

FedMut presents a novel federated learning framework for predicting combinatorial mutation effects that addresses key challenges in cross-institutional protein fitness prediction. By combining pre-trained protein language models with federated learning, we enable privacy-preserving collaborative training that can leverage diverse biological datasets while maintaining computational efficiency. Our approach demonstrates the potential for federated learning to advance protein engineering and evolutionary biology research by breaking down data silos and enabling collaborative model development.

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